St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Module: Musculoskeletal System | Lecture 8

# Upper Limb Cases


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# Objectives

| | |
|---|---|
| SOM.MKII.BPM1.1.MSK.1.ANAT.1199 | Describe the course of the radial, median, musculocutaneous, axillary and ulnar nerves in the arm, forearm and hand |
| SOM.MKII.BPM1.1.MSK.1.ANAT.1200 | Describe the signs and symptoms of an upper brachial plexus nerve lesion(s). |
| SOM.MKII.BPM1.1.MSK.1.ANAT.1201 | Describe the signs and symptoms of a lower brachial plexus nerve lesion(s). |
| SOM.MKII.BPM1.1.MSK.1.ANAT.1202 | Explain how a cervical rib may contribute to the development of a lower brachial plexus lesion |
| SOM.MKII.BPM1.1.MSK.1.ANAT.1203 | Describe the motor and sensory deficits of lesions of the radial nerve in the axilla, arm and wrist. |
| SOM.MKII.BPM1.1.MSK.1.ANAT.1204 | Explain why radial nerve injury at/above the elbow decreases grip strength. |
| SOM.MKII.BPM1.1.MSK.1.ANAT.1205 | Describe the signs and symptoms of lesions of the median nerve in the arm, wrist and proximal palm. |
| SOM.MKII.BPM1.1.MSK.1.ANAT.1206 | Explain the anatomical mechanism, signs and symptoms of “carpal tunnel syndrome” and the special test used for diagnosis. |
| SOM.MKII.BPM1.1.MSK.1.ANAT.1208 | Describe the signs and symptoms of lesions of the ulnar nerve in the arm and wrist and palm. |
| SOM.MKII.BPM1.1.MSK.1.ANAT.1162 | Explain the potential clinical consequences of fractures at the surgical neck, midshaft, supracondylar, and medial epicondylar regions of the humerus and be able to recognize them on a radiograph. |
| SOM.MKII.BPM1.1.MSK.1.ANAT.1182 | Identify the following carpal bone fractures: *scaphoid fracture* and fracture of hook of hamate and describe their possible clinical consequences. |
| *SOM.MKII.BPM1.1.MSK.1.ANAT.1210* | *Describe the functional tests for integrity of the median, ulnar and radial nerves in the forearm and hand (SG)* |
| *SOM.MKII.BPM1.1.MSK.1.ANAT.1211* | *Describe the biceps, triceps and brachialis reflexes and list the nerve roots being tested (SG)* |

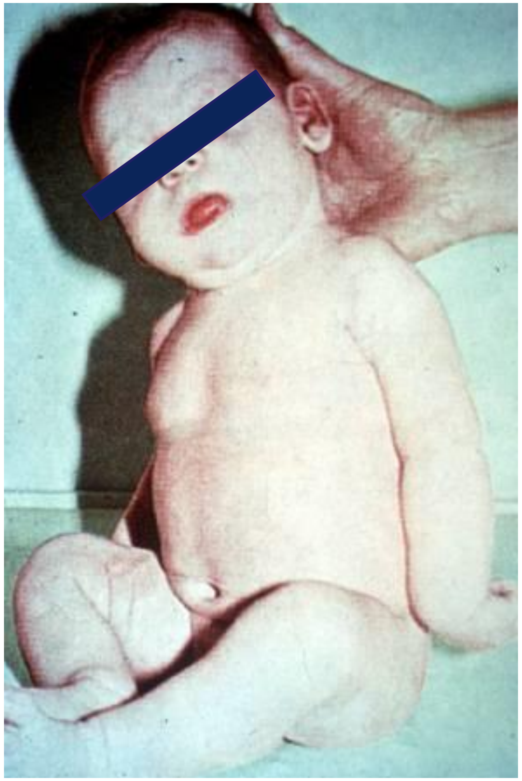
# Clinical Case 1

An 8-month-old boy was admitted to the pediatric department because of limited movement of his left upper limb since birth.

Physical examination show that his shoulder was internally rotated and adducted, his elbow was extended and forearm pronated.

What is the most likely injury?

- Joints affected - One or more than one?
- Muscle compartment affected - One or more than one?
- Nerve affected – one or more than one?
- If multiple nerves are involved, then the injury is likely to be traced back to a common origin: brachial plexus



# Clinical Case 1

Physical examination show that his shoulder was internally rotated and adducted, his elbow was extended and pronated.

**Shoulder – adducted and internally rotated**

Loss of – abduction Supraspinatus Deltoid Suprascapular (C5,6) Axillary (C5,6)

Loss of – external rotation Infraspinatus Teres minor Suprascapular (C5,6) Axillary (C5,6)

**Elbow – extended and pronated**

Loss of – flexion

Decreased - supination

Anterior compartment of arm Musculocutaneous (C5,6,7)

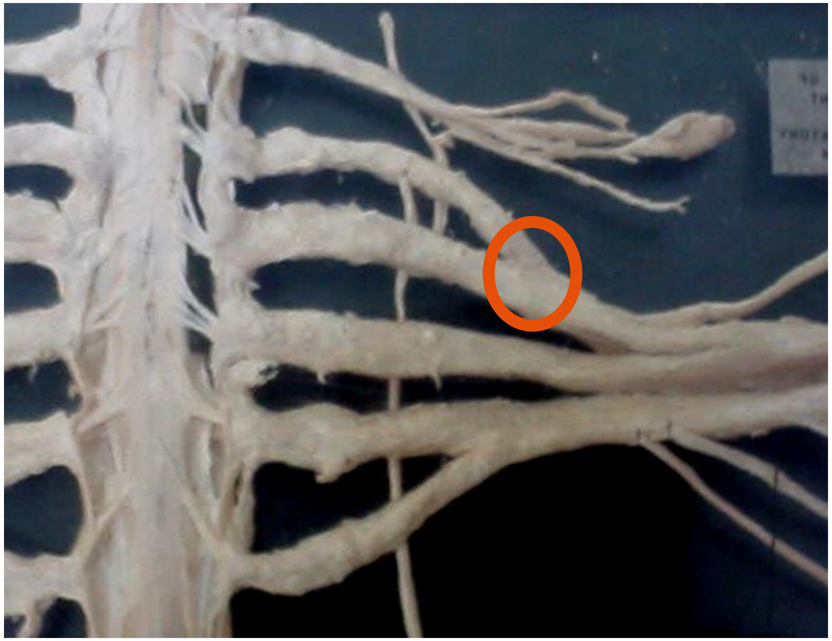
Do suprascapular, axillary and musculocutaneous come from the same nerves or parts of the brachial plexus? No

Injury is not to the individual nerves but higher (trunks and/or roots) Injury to upper brachial plexus lesion – C5, C6

# Upper Brachial Plexus Lesion: C5, 6 (7)

## Erb's Palsy

- Damage to upper parts of brachial plexus (roots &/or trunks)
- Arm adducted and internally rotated, forearm extended and pronated
- Clinically involved nerves are:
  - **Suprascapular:** inability to initiate abduction and loss of external rotation
  - **Axillary:** loss of abduction to 90 degrees and external rotation
  - **Musculocutaneous:** loss of forearm flexion and weakened forearm supination
- Sensory loss along the C5,6,+/-7 dermatomes – sergeant badge area, lateral border of the upper limb



Long thoracic nerve is spared if the injury is to the upper trunk

Long thoracic nerve can be involved in upper brachial plexus lesions if the injury involves proximal part of C5 and C6 nerve roots.

What will the patient present with?

# Discuss the clinical presentation of a lower brachial plexus injury

- Damage to lower parts of brachial plexus C8 / T1 lower trunk or medial cord
- Which nerves will be involved?
  - Ulnar FCU, med ½ FDP, most intrinsic muscles of hand, loss of sensation in medial hand
  - Med cut N of arm Loss of sensation along medial aspect of arm
  - Med cut N of forearm Loss of sensation along medial aspect of forearm
  - Med pectoral N
  - Medial head of Median LLOAF muscles

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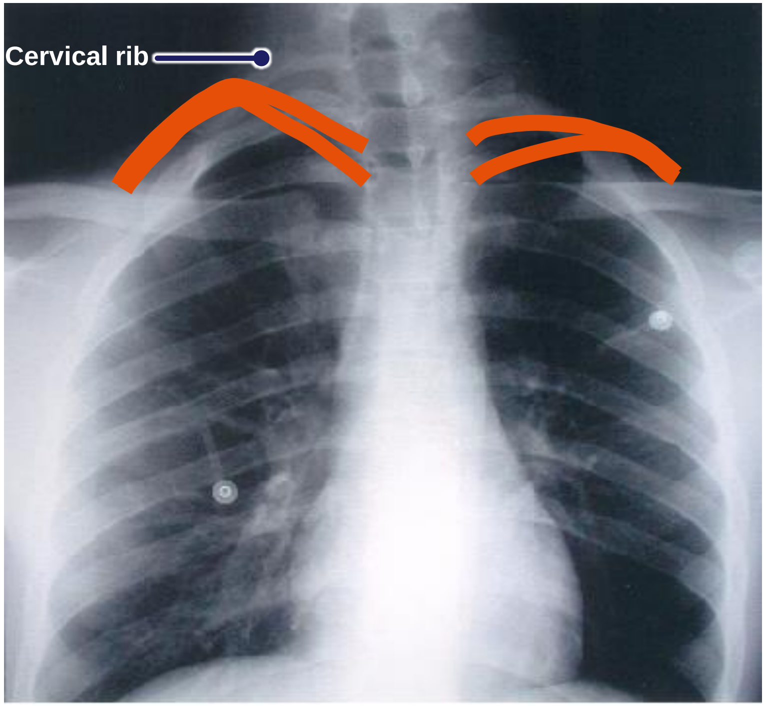
# Lower Brachial Plexus Lesion: C8, T1

- **Klumpke's (Klumpke-Dejerine) Palsy**
- Damage to lower parts of brachial plexus (trunks or medial cord)
- Loss of flexor carpi ulnaris (FCU) and medial ½ of flexor digitorum profundus (FDP)
- Loss of all intrinsic hand muscles (both ulnar and median)
- Sensory loss along C8 and T1 dermatomes - medial aspect of arm, forearm and hand




Cervical rib can be an underlying cause

# Clinical Case 2

A 27-year-old woman is brought to the emergency department after losing control during a motorbike race in which she was run over by several of the other racers. A radiograph of her right upper limb was taken, which is shown here.

- What is the most likely diagnosis?

**Midshaft fracture of the humerus**

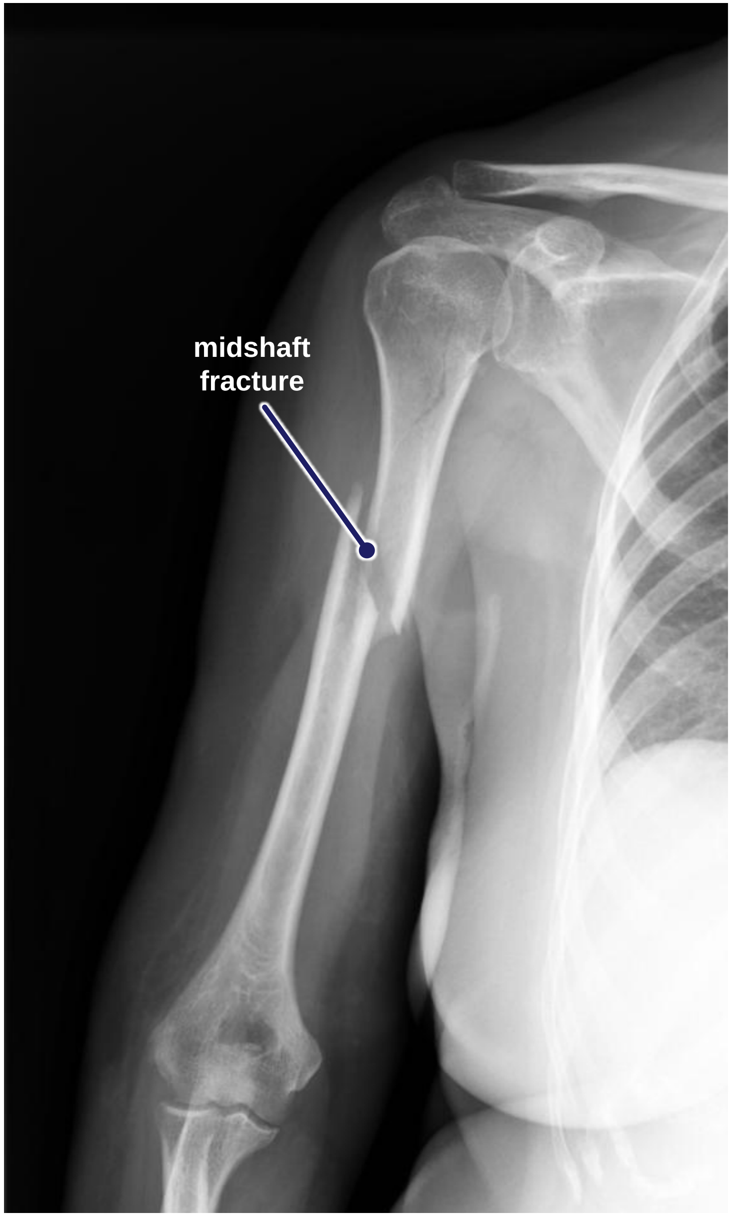
- Which structures are most likely damaged?

**Radial nerve and deep brachial artery**

- Why are they at risk?

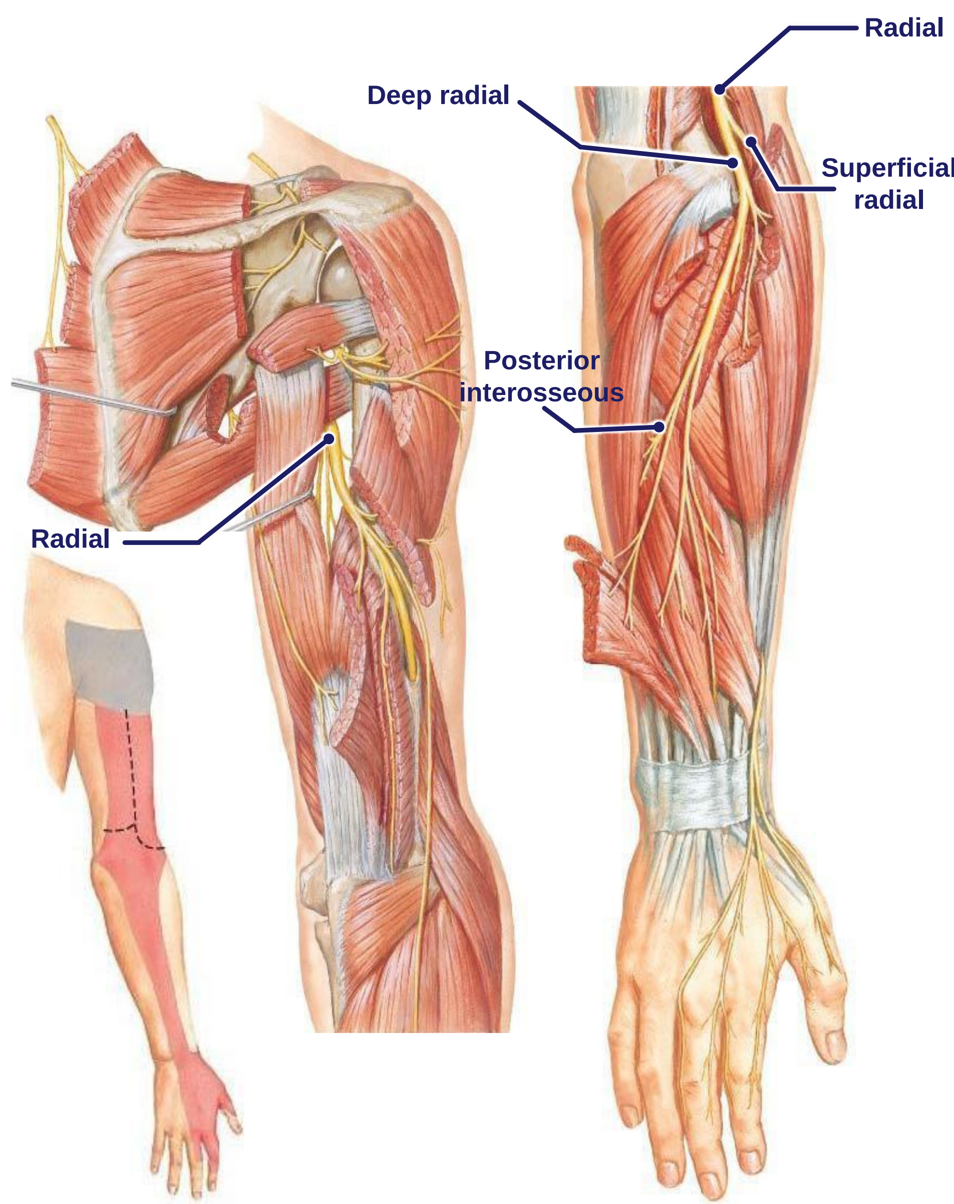
**Lie in radial groove**

- What motor and sensory deficits are most likely present?



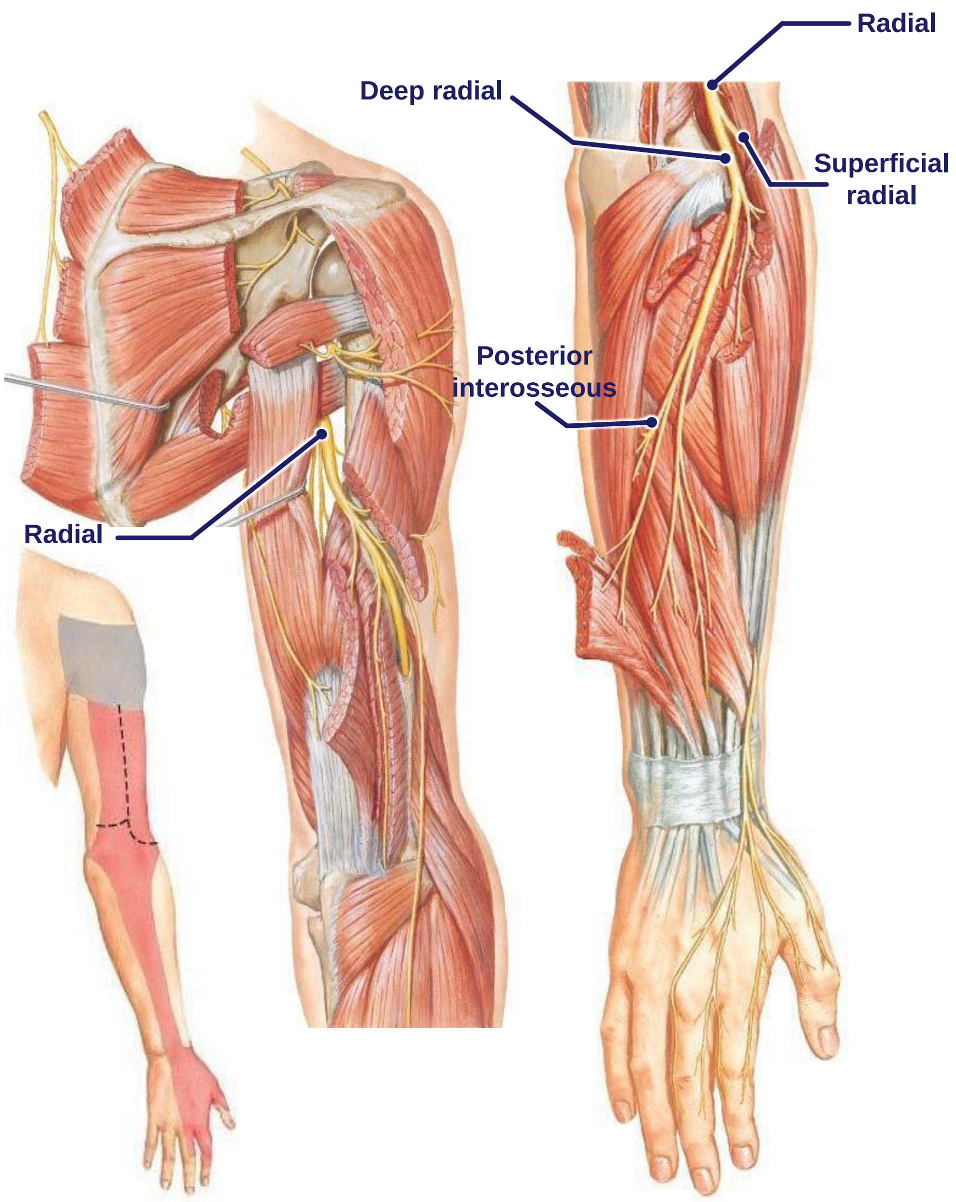
# Review: Summary of Pathway of Radial Nerve

- **Origin:**
  - From posterior cord
- **Course:**
  - Spiral (radial) groove of humerus
  - Gives sensory branches to the arm and forearm
  - Crosses elbow anterior to lateral epicondyle
  - Ends by splitting into **superficial radial (sensory)** and **deep radial (motor)** just distal to the elbow
  - **Deep radial** pierces the supinator and exits becoming the **posterior interosseous**
  - **Superficial radial** runs beneath brachioradialis in the forearm to emerge proximal to the styloid process of the radius



# Review: Summary of Pathway of Radial Nerve

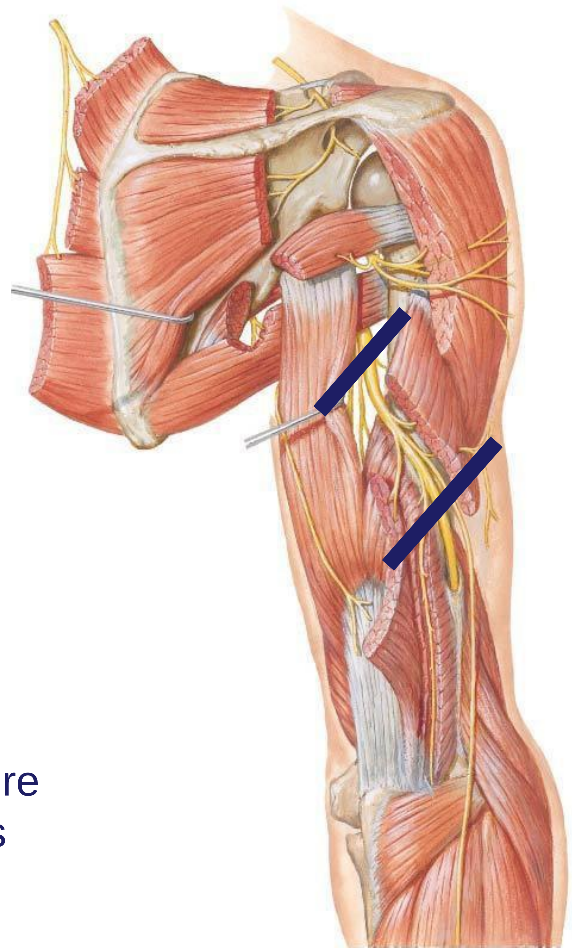
- **Motor supply:**
  - Supplies extensors of the forearm, wrist and digits (via the **radial n., deep radial n.** and **posterior interosseous n.**)
- **Sensory supply:**
  - **Posterior brachial cutaneous** – posterior arm
  - **Inferior lateral brachial cutaneous** – inferior and lateral arm
  - **Posterior antebrachial cutaneous** – posterior forearm
  - **Superficial radial** - dorsum of hand, lateral 2 ½ digits (proximal part of these digits)



# Common areas to injure the Radial Nerve

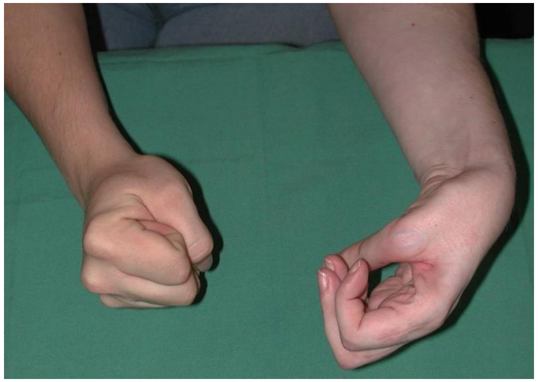
1. Injury in the Axilla e.g., Saturday night palsy

2. Injury in the arm e.g., midshaft fracture of the humerus



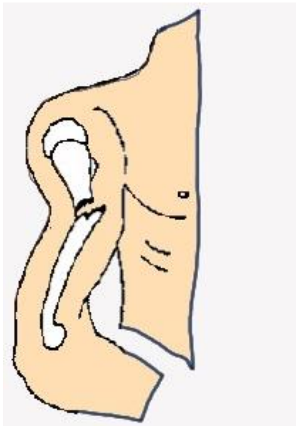
3. Injury in the forearm e.g., posterior interosseous nerve syndrome (motor only)

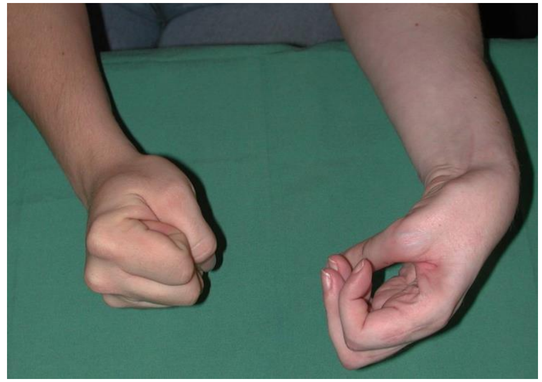
# Radial nerve injury in the axilla



- Loss of ability to extend the elbow joint.
- Loss of wrist extensors
  - Wrist drop
  - Impaired grip strength
    - Flexion of fingers are weakened due to the flexed wrist because the long flexor tendons cannot contract to their maximum ability
- Sensory loss:
  - Arm – posterior and lower lateral
  - Forearm – posterior
  - Hand – dorsum of hand – proximal part of lateral 2 ½ digits and associated dorsum of hand

# Radial nerve injury in the arm





- Retain ability to extend the elbow joint
  - The long and lateral heads of the triceps are supplied by the radial nerve before it enters the radial groove. Therefore, the muscle retains its function
- Loss of wrist extensors
  - Wrist drop
  - Impaired grip strength
- Sensory loss:
  - Arm - variable
  - Forearm – posterior
  - Hand – dorsum of hand – proximal part of lateral 2 ½ digits and associated dorsum of hand

A 27-year-old man is brought to the emergency department after losing control during a motorbike race in which he was run over by several of the other racers. Physical examination shows several cuts and bruises. He has difficulty extending his left wrist and left MCP joints and has difficulty in extending the IP joints of digits 2-5 of his left hand. He can extend the left elbow. There is no sensory loss. Which of the following nerves has most likely been injured to result in these signs, and in what part of the upper limb is the injury located?

- Are symptoms motor, sensory or both?
  - Motor
- Symptoms are associated with which joint/region
  - Wrist
  - Digits
- Motor symptoms are associated with which compartment
  - Loss of extension – posterior compartment
- Based on motor symptoms, which nerve(s)
  - Radial or its branches
- Based on motor symptoms, in which region is the nerve damaged – axilla, arm/forearm or hand?
  - Elbow extension is not affected – injury distal to triceps
  - Wrist extension is affected – injury proximal to wrist

  forearm
- Since patient has only motor symptoms, it can only be….
  - Radial
  - Sup radial
  - **Deep radial / post interosseous**

# Clinical Case 3

A 45-year-old woman comes to the emergency department because she fell on her outstretched hand. Physical examination shows tingling and decreased sensation on the thenar eminence and the lateral 3½ digits. Sensation is preserved in the mid palm.

- What is the most likely diagnosis?

**Lunate dislocation**

- What nerve is most likely injured and where?

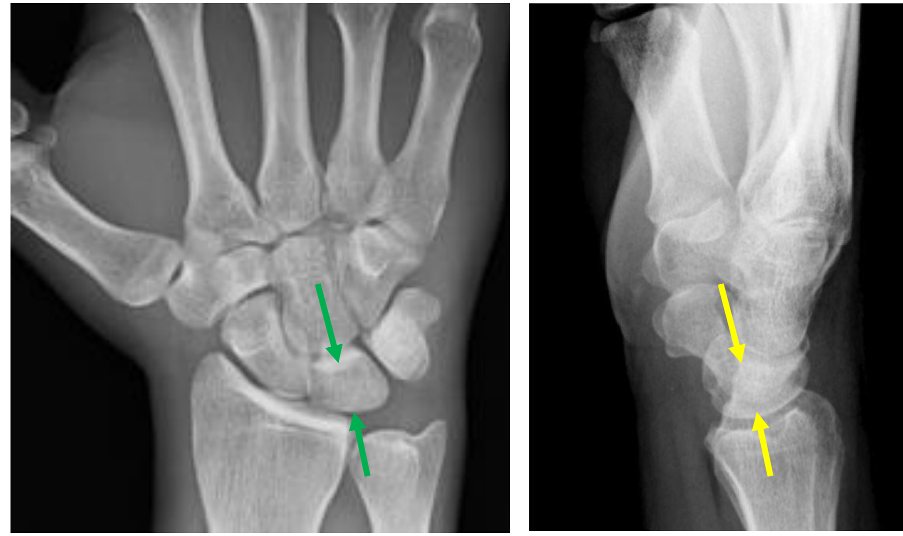
**Median in the Carpal tunnel**

- Why is sensation preserved in the mid palm?

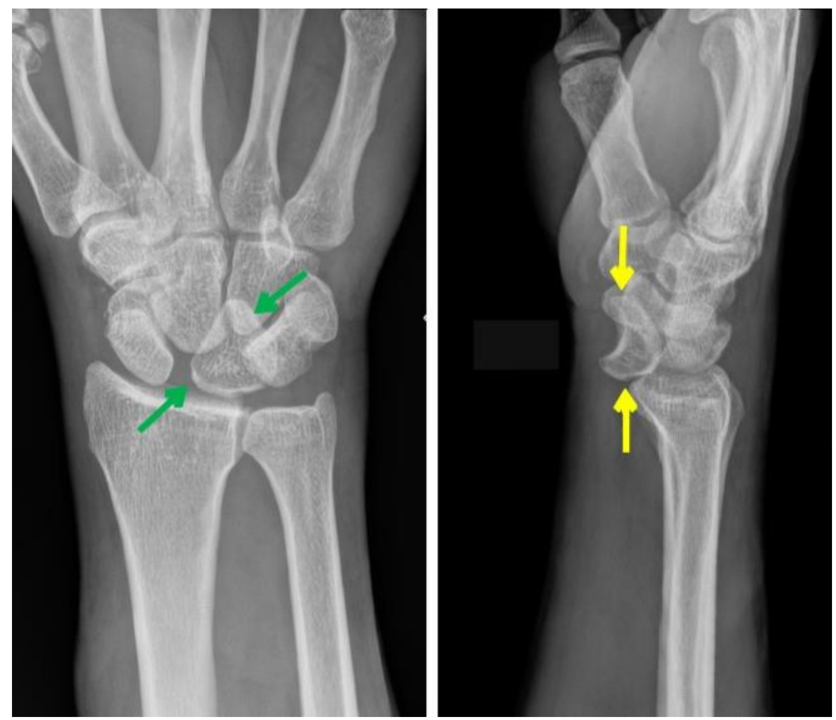
**Palmar cutaneous**

- What motor and sensory deficits are most likely present

**Normal Lateral Wrist**

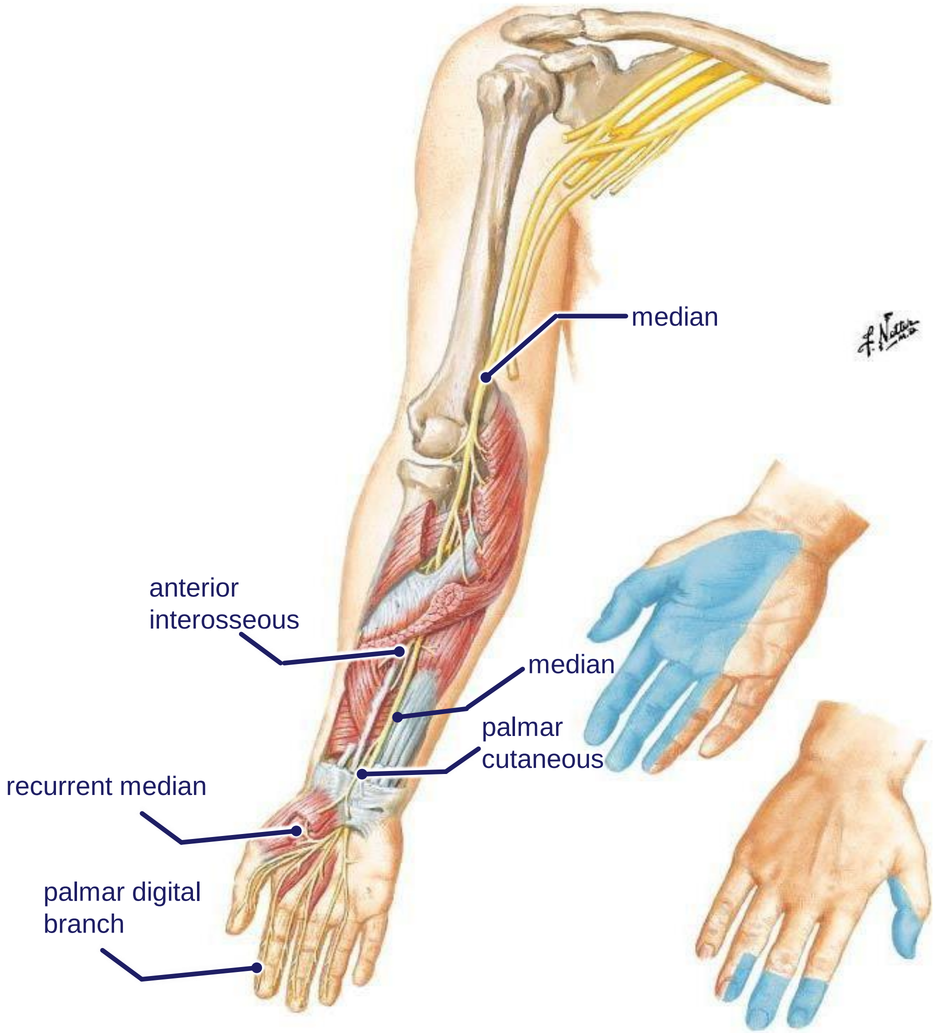


**Affected Patient**



# Review: Summary of Pathway of Median Nerve

- **Origin:** From lateral & medial cords
- **Course:**
  - Courses through the arm - no branches
  - Crosses anterior to elbow through cubital fossa
  - Just distal to the elbow, gives off the **anterior interosseous**
  - Gives **palmar cutaneous** branch in the distal forearm before
  - Passes through carpal tunnel into the hand
  - Gives **palmar digital branches** and **recurrent median**



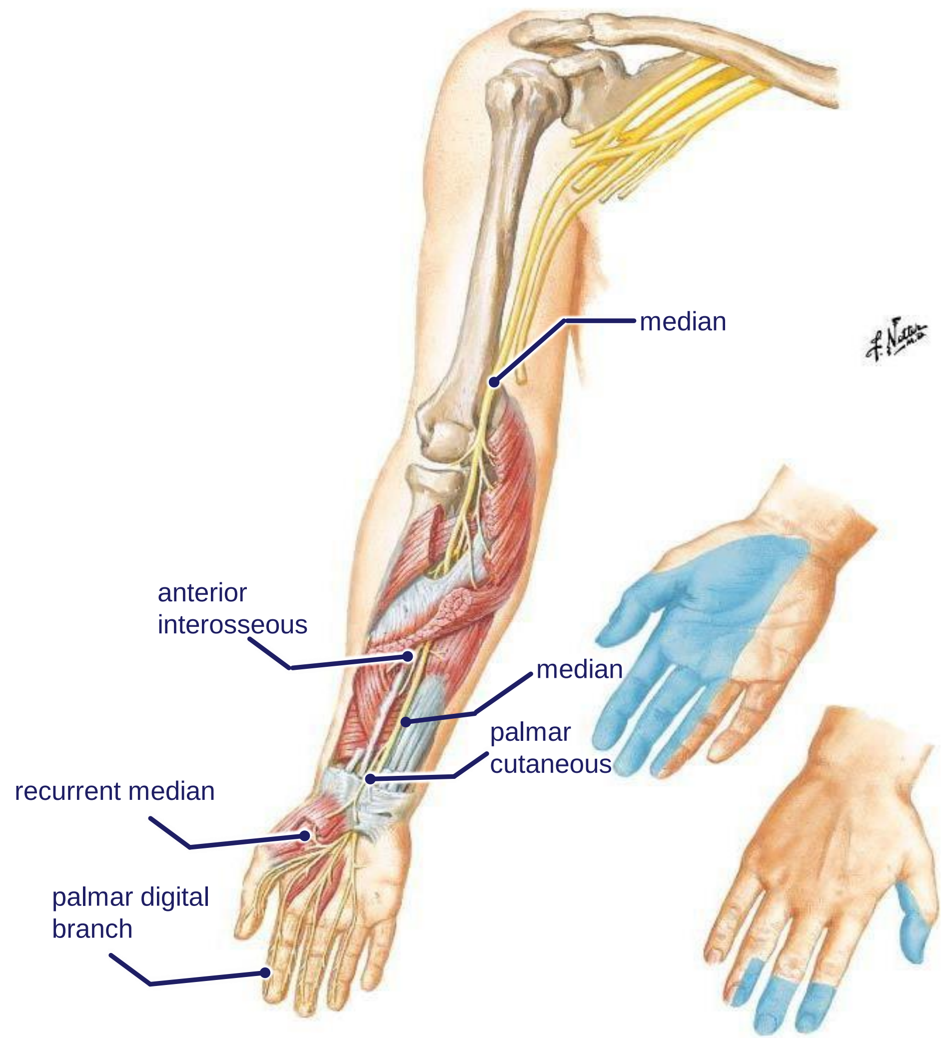
# Review: Summary of Pathway of Median Nerve

**Motor innervation:**

- Forearm:
  - Supplies all muscles in superficial and intermediate flexor compartment of forearm except FCU - **median**
  - Supplies the deep flexor compartment except ulnar half of FDP – **anterior interosseous**
- Hand:
  - 2 lumbricals (1st and 2nd) - **palmar digital branches**
  - OAF muscles - **recurrent branch**

**Sensory innervation:**

- **palmar digital branches** - lateral 3 ½ fingers palmar side and their distal phalanx on the dorsal side
- **palmar cutaneous branch** - mid-palm (comes off median in forearm and does not pass through carpal tunnel)



# Common areas to injure the Median Nerve

1. At/above the elbow e.g. supracondylar fracture

2. Injury to the anterior interosseous nerve(AIN) only e.g. entrapment of the AIN between the two heads of pronator teres

3. At the level of the wrist e.g. carpal tunnel syndrome, lunate dislocation, suicide attempt

4. Injury to the palm e.g. laceration to the base of the thumb

# Median nerve at/or above the elbow

**Elbow**

- Loss of pronation

**Wrist**

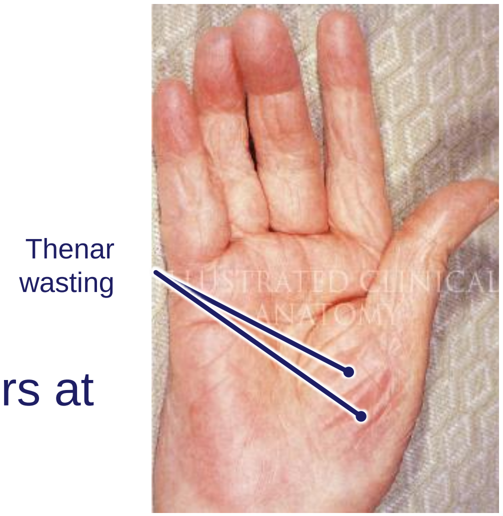
- Weakened wrist flexion
- Medial deviation (FCU unopposed)

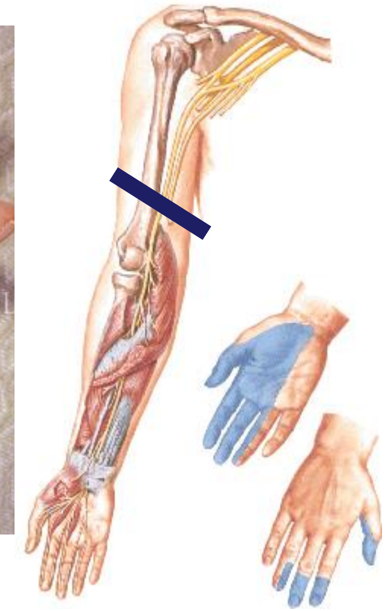
**Hand**

- Loss of flexion of index and middle fingers at DIP and PIP joints (FDP and FDS)
- Paralysis of 1st and 2nd lumbricals
- OAF muscles of thumb are paralyzed
- Loss of opposition of thumb
- Loss of flexion of the thumb
- Weakened abduction of thumb
- Thenar wasting due to atrophy of OAF muscles (later presentation)

**Sensory loss**

  - Lateral palmar 3 ½ digits to DIP on the dorsal aspect, associated surface of hand

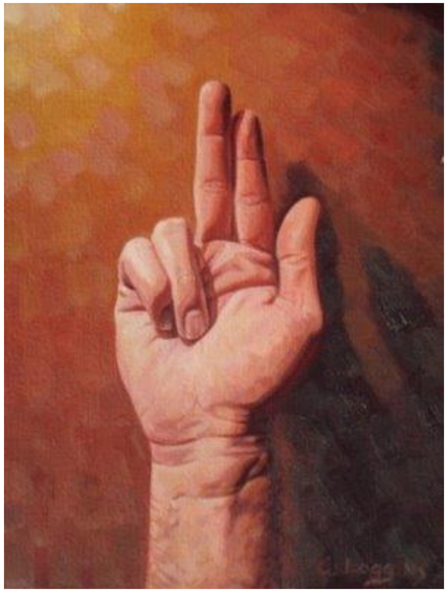


# Median nerve at/or above the elbow

**BENEDICTION HAND – occurs when patient tries to make a fist**

- The index and middle fingers stay straight (both FDS and FDP to these fingers have lost innervation)
- The ring and little fingers flex (FDP to these digits is intact – Ulnar)
- Thumb remains in plane of palm (flexor pollicis longus and brevis have lost innervation but adductor pollicis is intact – Ulnar)



# Median nerve injury at the anterior wrist

**Elbow and Wrist**

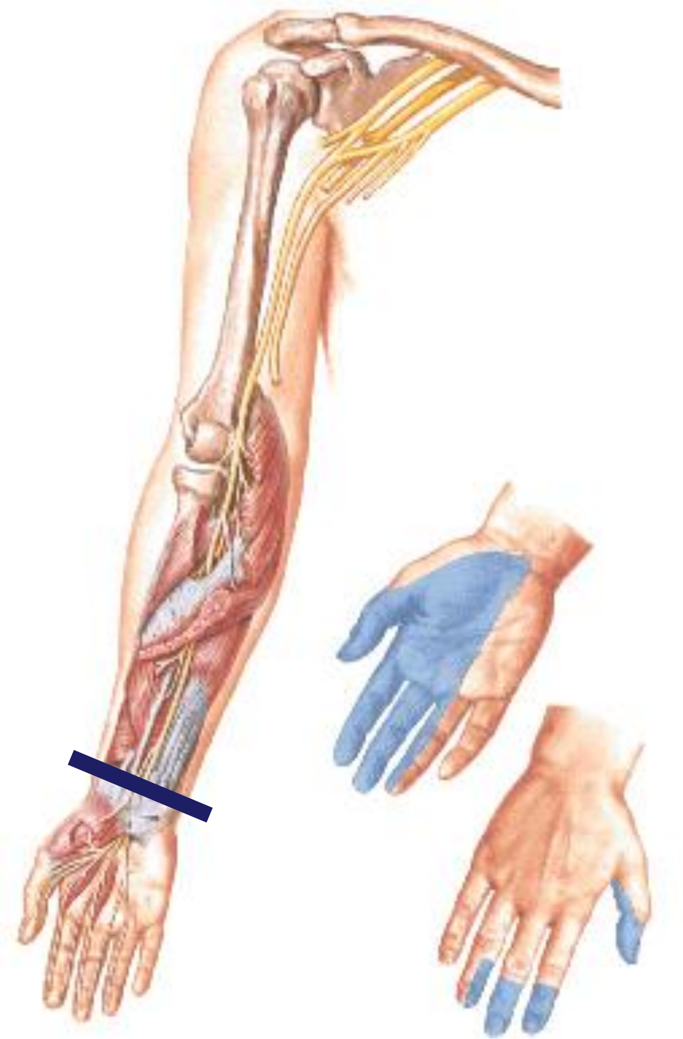
• Forearm pronation and wrist flexion intact, no medial deviation – No functional losses

**Hand**

- FDP and FDS intact - no benediction hand
- Paralysis of $1^{st}$ and $2^{nd}$ lumbricals
- OAF muscles of thumb are paralyzed
- Loss of opposition of thumb
- Weakened flexion and abduction of the thumb
  - Flexion and abduction are weakened but not lost because the flexor pollicis longus and abductor pollicis longus are still innervated
- Thenar wasting due to atrophy of OAF muscles (later presentation)

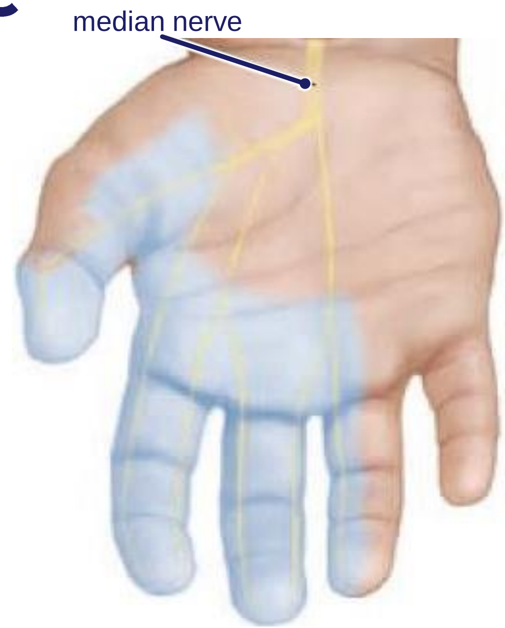
**Sensory loss**

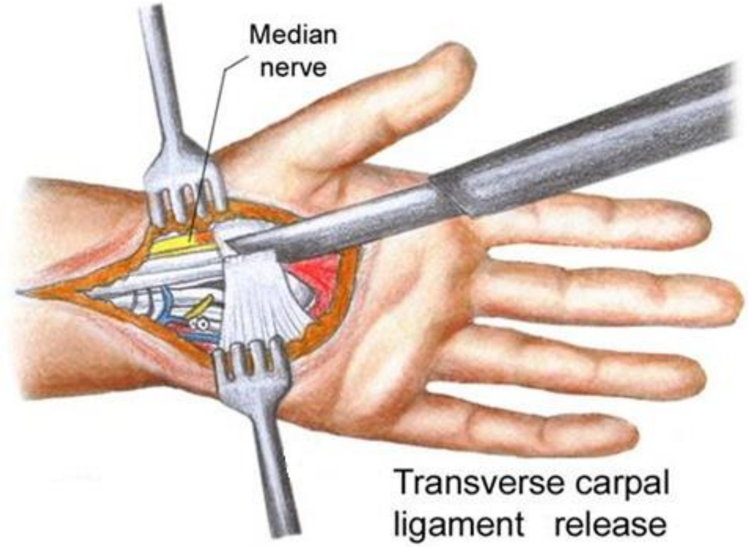
- Lateral palmar 3 ½ digits to DIP on the dorsal aspect, associated surface of hand



# Carpal Tunnel Syndrome

- Entrapment syndrome causing increased pressure in the carpal tunnel
- Compresses the median nerve
- Motor deficits: Loss of function to muscles supplied by median nerve in the hand - LLOAF (median nerve injury at the wrist)
- Sensory deficits: Paresthesia in the median nerve distribution in the hand (except central palm which is supplied by the palmar cutaneous branch), nocturnal pain common



# Tinel's Sign

# Phalen's Test

Sensory distribution of median n. after passing through carpal tunnel

**(A)** A positive **Tinel sign** refers to distally radiating pain and/or paresthesia elicited by percussing a superficial peripheral nerve, in this case the median.

**(B)** The **Phalen maneuver** is performed by apposing the wrists in flexion. Paresthesia in the hand within 60 seconds is considered a positive test.

# Median nerve injury at the proximal palm

## Recurrent branch - motor only nerve

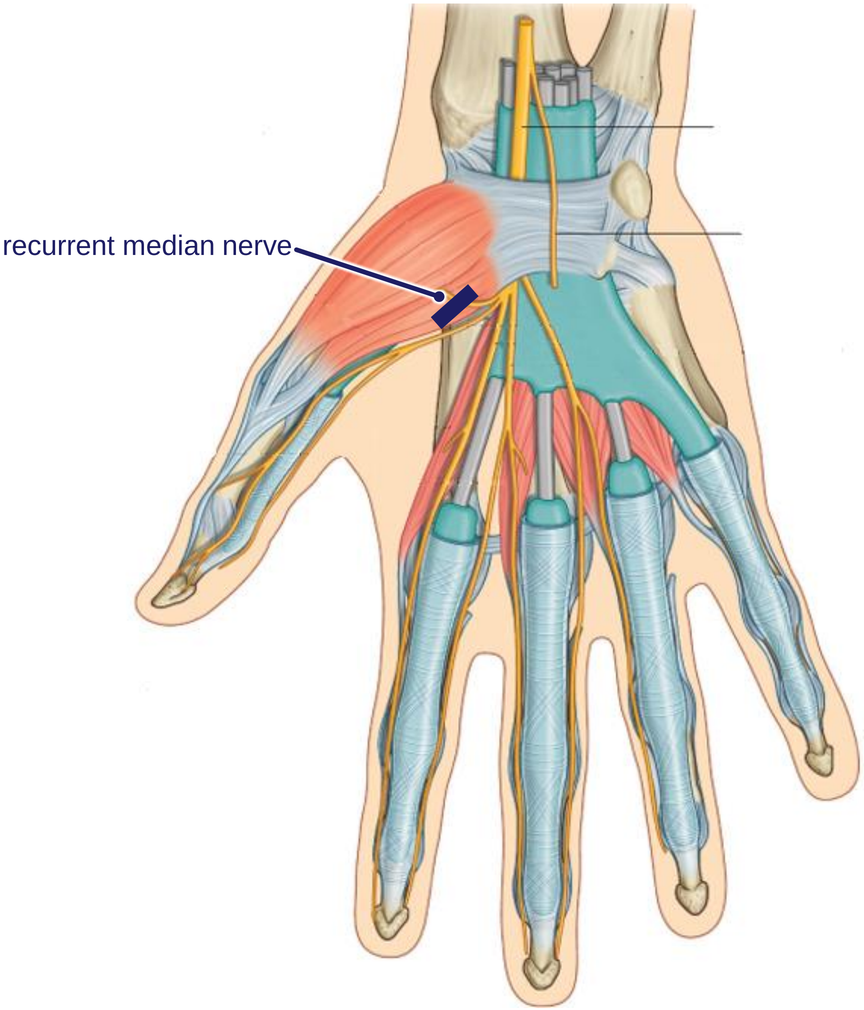
### Elbow and Wrist

- Forearm pronation and wrist flexion intact, no ulnar deviation – No functional losses

### Hand

- 1st and 2nd lumbricals **intact**
- Flexion of PIP and DIP **intact**
- OAF muscles of intrinsic hand are paralyzed – atrophy leading to thenar wasting (later presentation)
- Loss of opposition of thumb
- Weakened flexion of the thumb
- Weakened abduction of thumb

### No sensory loss



# Clinical Case 4

A 29-year-old golfer comes to the emergency department with a complaint of severe wrist pain that has increased in severity over the past few days. A radiograph and CT scan are taken which are shown.

- What is the bony injury?

**Fracture of the hook of the hamate**

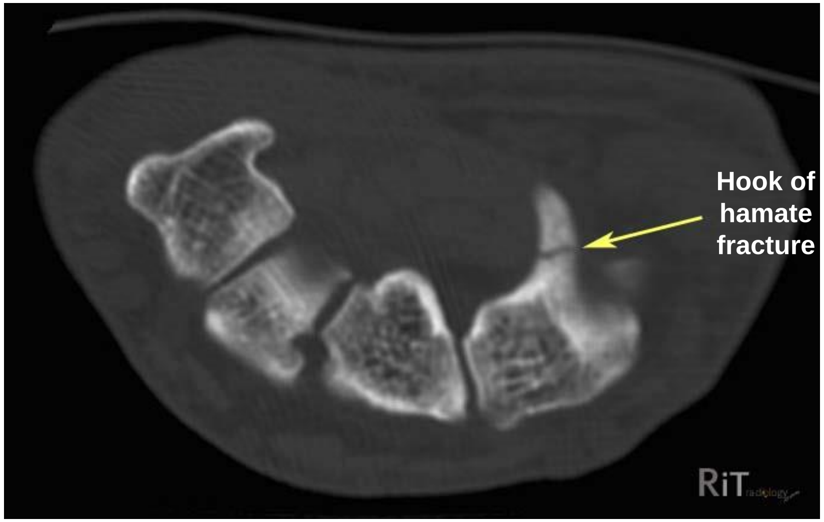
- Which neurovascular structures are at most risk?

**Ulnar nerve and vessels**

- Where are they at risk?

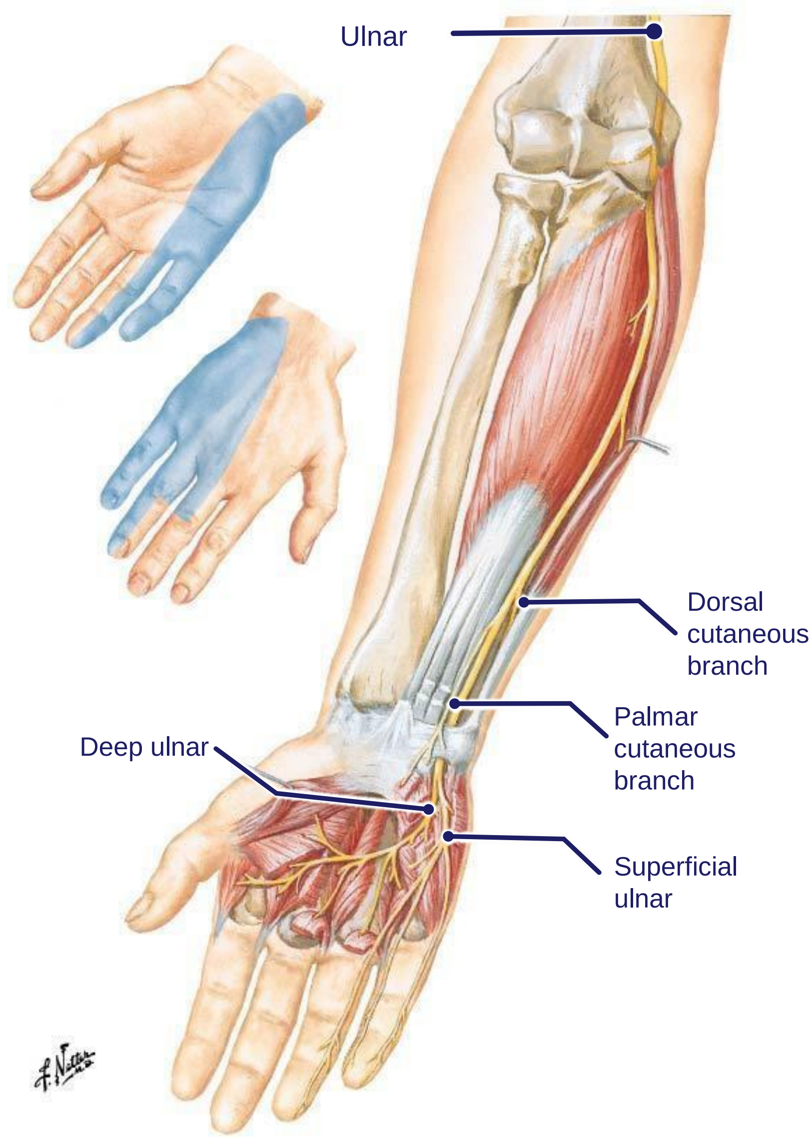
**Guyon's canal**

- What motor and sensory deficits are most likely present

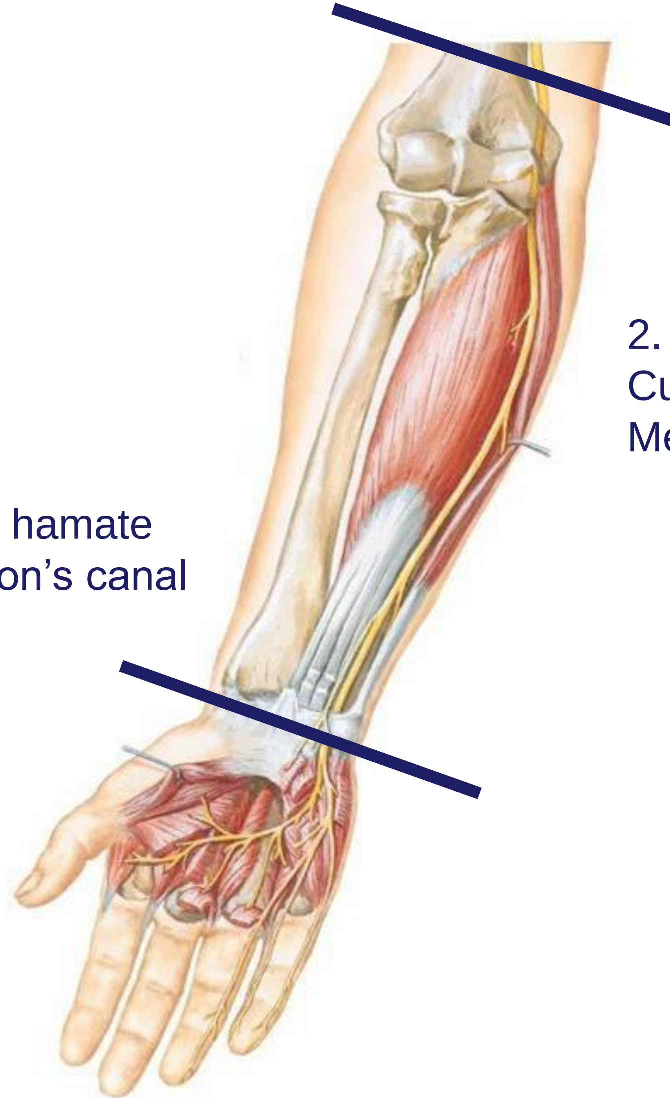


# Summary of Pathway of Ulnar Nerve

- **Origin:** From medial cord
- **Course:**
  - Courses through the arm - no branches
  - Crosses behind medial epicondyle
  - Descends forearm under the flexor carpi ulnaris muscle
  - At distal third of the forearm gives a **dorsal cutaneous branch**
  - Crosses the wrist superficial to flexor retinaculum at Guyon's canal
  - Divides into **superficial ulnar (sensory)** and **deep ulnar (motor)**
- **Motor innervation**
  - Supplies flexor carpi ulnaris and medial half of flexor digitorum profundus - **ulnar**
  - Motor branch to all the muscles of the hand except LLOAF - **deep ulnar**
- **Sensory innervation**
  - **Superficial ulnar** - sensory to the palmar medial 1 ½ digits
  - **Dorsal cutaneous branch** - sensory to the dorsal medial 2 ½ digits



# Common areas to injure the Ulnar Nerve



1. At the wrist:
Suicide attempt,
Fracture of hook of hamate
Entrapment in Guyon's canal

2. At the elbow:
Cubital tunnel syndrome,
Medial epicondyle fracture

# Ulnar nerve injury at/or above the elbow

**Wrist**

- Weakness of wrist flexion– paralysis of FCU and medial FDP
- Lateral deviation of wrist – paralysis of FCU

**Hand**

- Lose flexion of ring & little fingers at DIP (FDP paralysis)
- Paralysis of hypothenar muscles
- Paralysis of $3^{rd}$ and $4^{th}$ lumbricals: decreased flexion of MCP, decreased extension at IP joints of ring and little finger
- Paralysis of all interosseous muscles –loss of abduction and adduction of digits
- Paralysis of adductor pollicis - loss of adduction of thumb

**Sensory loss**

- Medial palmar 1 ½ & dorsal 2 ½ fingers, associated surfaces of hand

# Ulnar nerve injury at the anterior wrist

## Wrist

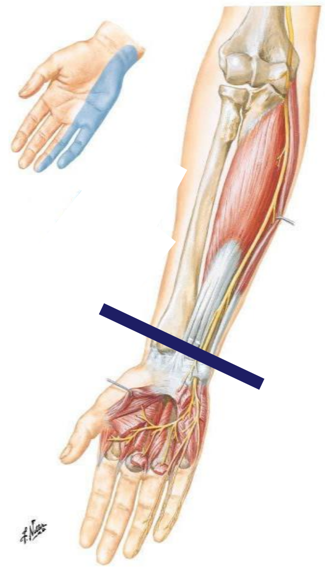
- Wrist flexion intact, no lateral deviation (FCU and ulnar half of FDP intact) – No functional losses

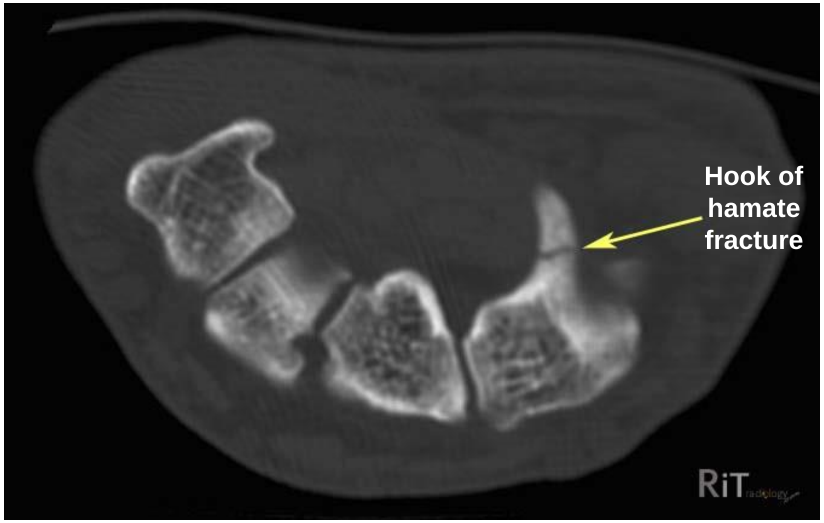
## Hand

- Paralysis of hypothenar muscles
- Paralysis of 3rd and 4th lumbricals: decreased flexion of at MCP – leads to hyperextension, decreased extension at IP joints of ring and little finger
- Paralysis of all interosseous muscles – loss of abduction and adduction of digits (metacarpal guttering)
- Paralysis of adductor pollicis - loss of adduction of thumb

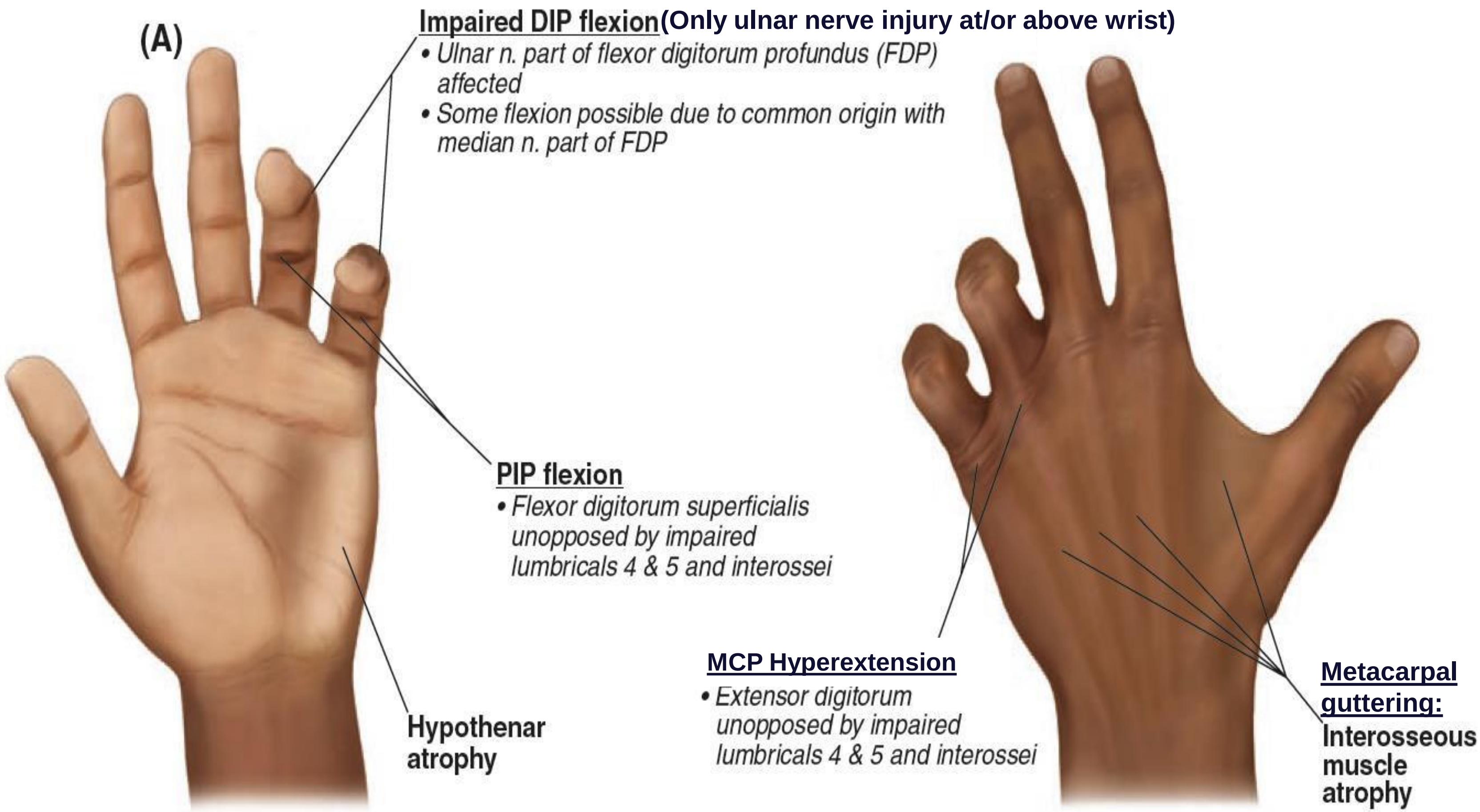
## Sensory loss

- Medial palmar 1 ½ fingers, associated surface of hand



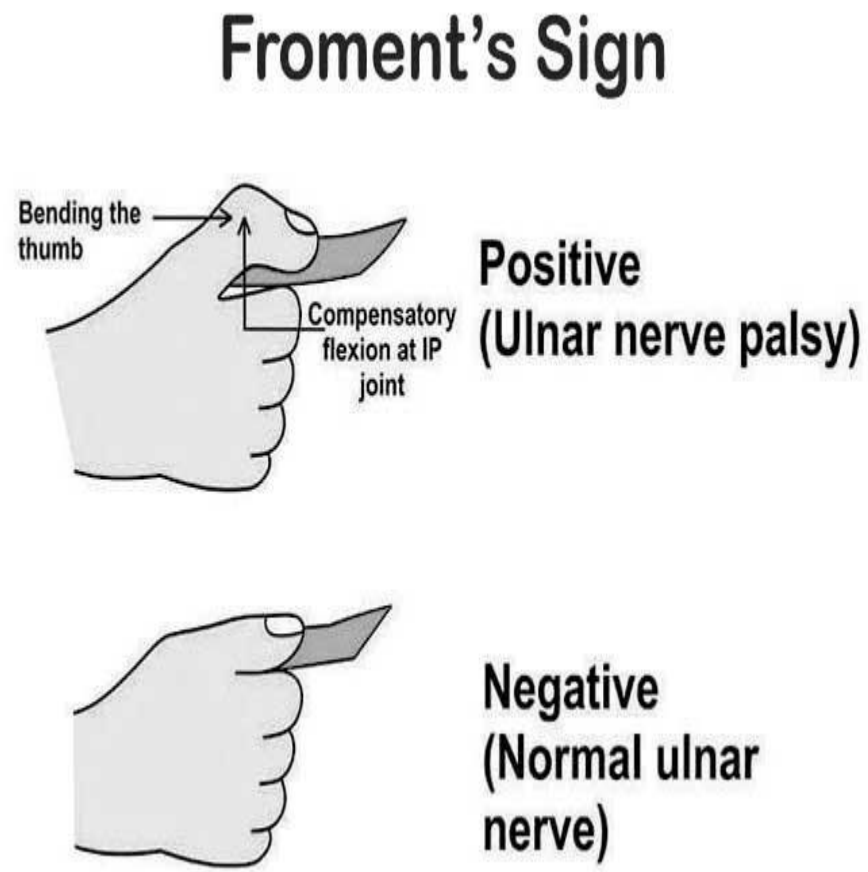


# Ulnar nerve Injury



# Froment's Test

- Tests the adductor pollicis muscle
- Place a sheet of paper between the thumb and index finger of the hand to be tested
- Ask the patient to resist as you try to pull on the sheet without 'bending' their thumb
- Patient can't hold on to the paper with adductor muscle, but FPL and FPB are intact (median) which allows for "holding on" by FLEXING the interphalangeal joint of the thumb




https://physio-study.com/book-test-for-ulnar-nerve-forments-sign/

# Clinical Case 5

A 5-year-old girl was brought to the emergency after falling from a tree. She was in severe pain and had limited mobility of the shoulder. A radiograph was taken, which is shown here.


- What is the bony injury?

**Fracture of the surgical neck of the humerus**

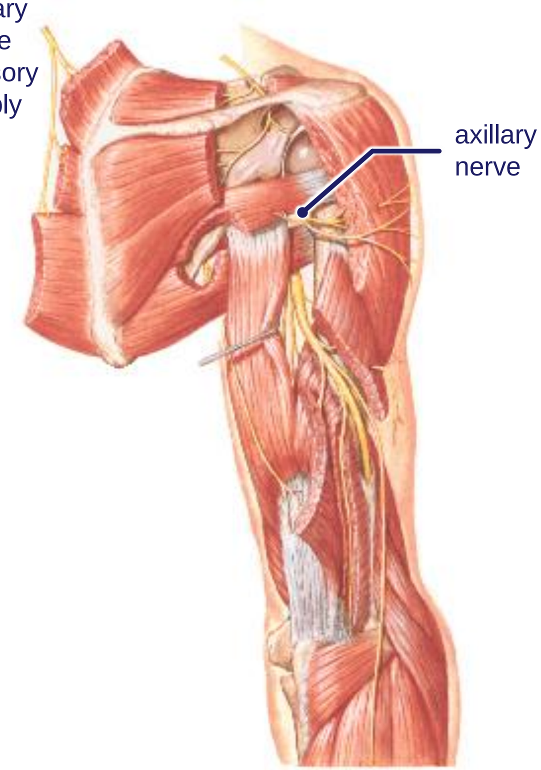
- Which neurovascular structures are most likely damaged?

**Axillary nerve and posterior humeral circumflex artery**

- What motor and sensory deficits are most likely present

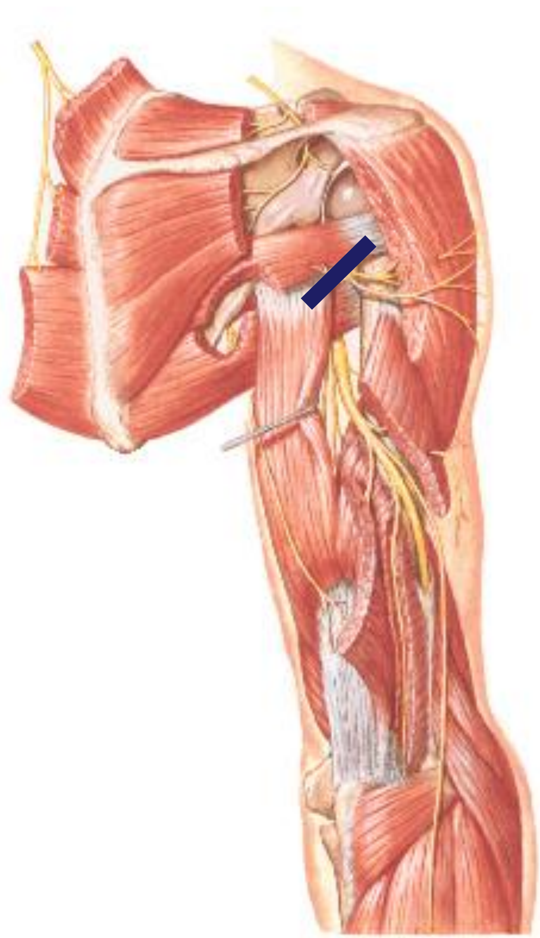
# Review: Summary of Pathway of Axillary Nerve

- **Origin:**
  - From posterior cord
- **Course:**
  - Passes inferiorly and laterally to exit through the quadrangular space accompanied by the posterior humeral circumflex artery
  - Passes posterior to the surgical neck of humerus
- **Motor supply:**
  - Supplies deltoid and teres minor
- **Sensory supply:**
  - Upper lateral part of arm

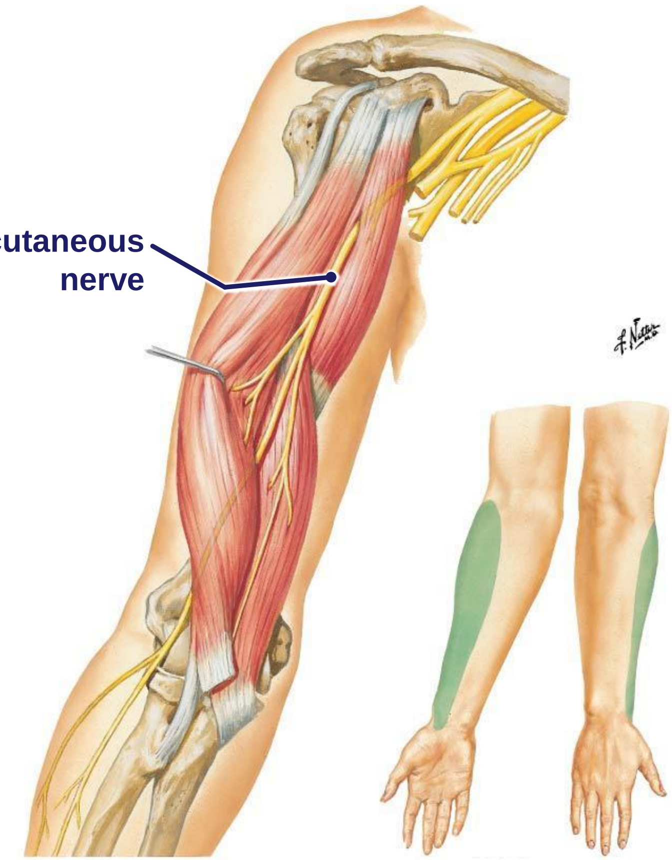
# Axillary nerve injury

- **Arm**
  - Decreased lateral rotation
  - Decreased abduction of arm
- **Sensory loss**
  - Loss of sensation to the lateral shoulder (regimental patch)



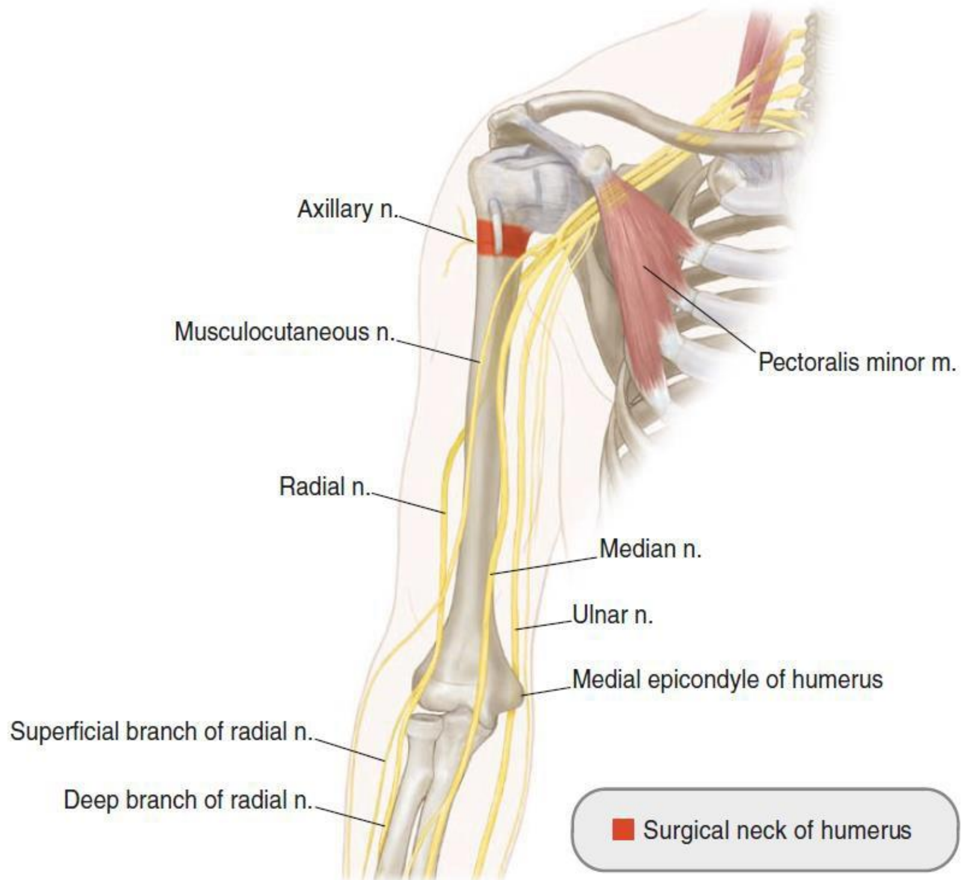
# Homework!

- Origin?
- Course?
- Terminates?
- Innervation?
- Mechanism of injury?
- Motor loss?
- Sensory loss?



# Review your humeral fractures!!

- Surgical neck
- Mid-shaft
- Supracondylar
- Medial epicondyle



# Work Sheet!!

| Lesion | *Major* Nerves affected | *Major* Muscles affected | Sensory and motor loss |
|---|---|---|---|
| Upper brachial plexus lesion<br>• Erb's palsy<br>• C5 & C6 nerve roots or upper trunk | | | |
| Lower brachial plexus lesion<br>• Klumpke's palsy<br>• C8 & T1 nerve roots or lower trunk | | | |

# Work Sheet!!

| Damage | Motor Deficits | Sensory Deficits |
|---|---|---|
| RADIAL NERVE | | |
| Axilla | | |
| Radial groove | | |
| Posterior interosseous nerve | | |
| Superficial branch of the radial nerve | | |

# Work Sheet!!

| Damage | Motor Deficits | Sensory Deficits |
|---|---|---|
| MEDIAN NERVE | | |
| proximal to or/at the elbow | | |
| anterior interosseous nerve | | |
| at the wrist | | |
| recurrent motor branch | | |

# Work Sheet!!

| Damage | Motor Deficits | Sensory Deficits |
|---|---|---|
| ULNAR NERVE | | |
| Arm or medial epicondyle | | |
| Damage at the wrist at Guyon canal | | |
| Deep branch of the ulnar | | |
| Superficial branch of the ulnar nerve | | |